

Comment on the Development of an Artificial Intelligence Action Plan (Docket Number: NSF_FRDOC_0001-3479)

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Good day Mr. D'Souza,

The United States has led the world in artificial intelligence innovation, not just because of our technological capabilities, but because of our commitment to fostering free enterprise, securing national interests, and maintaining strategic superiority in emerging domains. AI is now an integral part of global competition, influencing military strategy, economic stability, and national security. As the federal government moves to define an *AI Action Plan* under the direction of President Trump's [Executive Order on Removing Barriers to American Leadership in AI](#), I appreciate the opportunity to provide input on the priority areas that will shape the future of AI in America.

The choices made today will not only define U.S. AI leadership in the coming decade but will also set a global precedent for AI's role in national security, economic independence, and ethical governance. A proactive, security-conscious, and innovation-driven *AI Action Plan* is not merely an advantage -- it is a necessity for safeguarding the future of American strength and leadership in an AI-dominated world.

Therefore, the path forward must recognize AI as both an opportunity and a responsibility -- one that requires the United States to lead not only in innovation but also in cybersecurity, military applications, and responsible governance to prevent adversarial misuse. To this end, I offer three key recommendations for consideration:

1. AI for Cyber Defense and National Security

Artificial intelligence is fundamentally transforming modern warfare and cybersecurity. The United States must prioritize AI integration to maintain its strategic edge and protect national interests. As an incoming U.S. Army Cyber Warfare Officer, I recognize the critical importance of embedding AI into both defensive and offensive cyber operations. To this end, I propose the following priorities:

- **Strengthening AI-Driven Cyber Threat Detection:** The complexity and volume of cyber threats necessitate advanced AI solutions capable of real-time threat detection and

response. The Cybersecurity and Infrastructure Security Agency has developed a [Roadmap for Artificial Intelligence](#), aiming to enhance cybersecurity capabilities through AI adoption. This roadmap outlines efforts to promote beneficial uses of AI, ensure AI systems are secure, and deter malicious AI applications that could threaten critical infrastructure.

- **Enhancing AI Wargaming for Cyber Personnel:** Simulated environments powered by AI are essential for training cyber personnel in real-time threat scenarios. The Department of Defense has launched the [Artificial Intelligence Rapid Capabilities Cell](#) to accelerate the adoption of advanced AI technologies. This initiative includes pilot programs that apply generative AI models to both warfighting and enterprise management, providing immersive training platforms for cyber operators.
- **Expanding AI Research Collaborations Between the Department of Defense and Private AI Firms:** Public-private partnerships are vital for keeping national security applications at the forefront of technological advancement. The DoD's Chief Digital and Artificial Intelligence Office has initiated collaborations with private sector entities to integrate cutting-edge AI solutions into defense operations. These partnerships aim to leverage commercial innovation to enhance military capabilities, ensuring that the U.S. maintains a competitive advantage in AI applications for national security.

Bottom line: The United States must integrate AI-driven threat detection, immersive AI training for cyber personnel, and robust public-private AI research collaborations to enhance national security and protect critical infrastructure.

2. AI Policy Must Align with U.S. Strategic Interests

As artificial intelligence continues to evolve, it is imperative that U.S. AI policy not only fosters innovation but also aligns seamlessly with national security objectives. While streamlining bureaucratic processes can accelerate technological advancements, a robust governance framework is essential to safeguard against potential threats posed by adversarial entities. To this end, the following measures are recommended:

- **Preventing Exploitation of U.S.-Developed AI Technologies:** The global proliferation of AI technologies necessitates stringent controls to prevent adversarial nations from leveraging U.S. innovations for malicious purposes, such as cyber warfare or economic manipulation. The National Security Commission on Artificial Intelligence had [emphasized in its final report](#) the urgency of protecting AI assets, noting that the U.S. is not adequately prepared to defend or compete in the AI era, particularly against nations like China that are rapidly advancing their AI capabilities. To mitigate these risks, the U.S. must implement comprehensive export controls and safeguard intellectual property to ensure that AI technologies do not fall into the hands of adversaries.
- **Establishing Security Compliance Standards in Critical Sectors:** AI applications are increasingly integrated into critical sectors, including defense, law enforcement, and intelligence operations. To maintain national security, it is essential to develop and enforce security compliance standards tailored to AI systems within these domains. The previous administration's [National Security Memorandum on Artificial Intelligence](#), issued in October 2024, provided directives for federal agencies, aiming to ensure that AI adoption enhances national security while adhering to safety and ethical standards. Though the President revoked this memorandum last month, the need for standardized evaluations and policies to test, understand, and mitigate risks associated with AI systems before their deployment in sensitive areas is in no way reduced.
- **Addressing Risks of AI-Generated Misinformation and Deepfakes:** The advent of AI-generated content, particularly deepfakes, poses significant threats to national stability by enabling the spread of misinformation and facilitating foreign influence campaigns. The National Security Agency and other federal agencies [have highlighted the cybersecurity challenges](#) presented by “synthetic media”, which can be used to manipulate public opinion or undermine trust in institutions. To combat these threats, it is crucial to develop advanced detection technologies, establish legal frameworks to deter malicious use, and promote public awareness about the potential dangers of deepfakes.

Bottom line: Federal policy must implement stringent export controls, enforce security compliance in critical sectors, and combat AI-generated misinformation to prevent adversarial exploitation and safeguard democratic institutions.

3. AI for Economic and Workforce Development

Maintaining American leadership in AI necessitates a robust domestic workforce proficient in AI-related disciplines, particularly in cybersecurity, defense applications, and AI law. Federal AI policy should focus on the following strategic areas:

- **Investing in AI Training for Military and Cybersecurity Personnel:** Equipping military and cybersecurity professionals with advanced AI skills is crucial for national security. Initiatives like the [Cybersecurity and Artificial Intelligence Talent Initiative](#) aim to recruit and train a world-class digital workforce by placing recent graduates in federal agencies with cybersecurity needs, providing them with vital public sector experience and leadership development opportunities. Additionally, educational institutions are stepping up; for instance, Sam Houston State University [has launched SamPoly](#), offering online certificates and associate degrees in practical AI and intelligent automation to address workforce shortages in Texas.
- **Supporting AI-Driven Domestic Manufacturing and Chip Production:** Reducing reliance on foreign supply chains for semiconductors is vital for AI advancement. The CHIPS and Science Act of 2022 allocates substantial funding to boost domestic semiconductor manufacturing. Building on this, the [Building Chips in America Act of 2023](#) streamlines approvals for semiconductor projects by exempting them from certain environmental reviews, expediting the establishment of manufacturing facilities. These legislative efforts aim to enhance national security and supply chain resilience by fostering a robust domestic semiconductor industry.
- **Encouraging AI Innovation with Security-Focused Safeguards:** Promoting AI innovation requires balancing technological advancement with the protection of intellectual property. The previous Administration [proposed a regulatory framework](#) to ensure the responsible diffusion of frontier AI technology globally, aiming to preserve U.S. leadership in AI while safeguarding national security interests. This framework sought to control the export of advanced AI chips and related technologies, preventing adversarial nations from exploiting U.S. innovations for malicious purposes.

Bottom line: The Federal Government must invest in workforce development for cybersecurity and military personnel, strengthen domestic semiconductor manufacturing to reduce foreign dependency, and promote AI innovation while enforcing safeguards to protect national security and intellectual property.

As the United States moves forward in shaping its *Artificial Intelligence Action Plan*, the policies implemented today will define the nation's ability to secure its technological future, maintain strategic superiority, and defend against adversarial threats. AI is not merely an economic tool but a critical pillar of national security, military readiness, and economic resilience. By prioritizing AI-driven cybersecurity defenses, aligning AI governance with national security objectives, and ensuring robust workforce development, the U.S. can solidify its leadership in this rapidly evolving field.

Furthermore, fostering public-private partnerships, safeguarding domestic AI innovation from foreign exploitation, and investing in domestic semiconductor production will not only strengthen American economic independence but also reinforce U.S. defense capabilities in cyberspace and beyond. These efforts must be pursued without compromising national security safeguards or allowing foreign adversaries to exploit American advancements in AI.

With AI reshaping modern warfare, global competition, and critical infrastructure, the United States cannot afford to take a reactive approach. The *AI Action Plan* must be decisive, forward-looking, and unwavering in its commitment to strengthening national security, enhancing economic resilience, and empowering the next generation of AI professionals. I urge the National Science Foundation and relevant agencies to incorporate these recommendations and ensure that American AI policy reflects the urgency of this moment.

Thank you for your time and consideration. I look forward to further discussions on this topic and welcome any opportunities to contribute insights as the AI Action Plan takes shape.

Yours respectfully,

Mateo Portelli

Further Reading

To provide additional context and inform the development of the *AI Action Plan*, the following resources offer comprehensive insights into the intersection of artificial intelligence and national security:

- [Department of Homeland Security Artificial Intelligence Roadmap](#): This 2024 document outlines DHS's plans for AI integration across its operations, including initiatives to bolster AI-driven cybersecurity and strategies to mitigate AI-related risks.
- [“The Real-World Impact of AI on Cybersecurity Professionals” from \(ISC\)²](#): This 2024 article explores how AI is influencing the roles of cybersecurity professionals, enhancing job efficiency, and transforming job functions within the industry.
- [“How AI Might Affect Decisionmaking in a National Security Crisis” from the Carnegie Endowment for International Peace](#): This 2024 article delves into the potential impacts of AI on decision-making processes during national security crises, considering both the benefits and challenges.
- [“Knowledge-Integrated Informed AI for National Security” report](#): This 2022 paper out of Cornell University discusses the integration of domain-specific knowledge into AI systems to enhance their effectiveness and reliability in national security applications.
- [“How Artificial Intelligence Is Transforming National Security” from the U.S. Government Accountability Office](#): This 2022 blog post examines how AI is expected to revolutionize various sectors, including national defense, and the Department of Defense's investments in AI integration.
- [“Artificial Intelligence and National Security” from the Bipartisan Policy Center](#): This 2020 brief discusses the strategic importance of AI in national defense and the necessity for the United States to prepare for AI's national security implications.